ASLAM SIKDER

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PROFESSIONAL SUMMARY

Results-driven AI Engineer, Researcher and Educator with around 5 years of experience in academy and industry. Proven expertise in Edge AI, Agentic AI, RAG, NLP, Gen AI, on device LLM and SLM, and ensemble modeling. Strong academic background with multiple peer-reviewed publications, focused on predictive modelling. Passionate about building real-world AI solutions and bridging research with practical impact. I aim to utilize my strong theoretical foundation and technical expertise in this field to design and develop scalable, intelligent solutions.

RESEARCH INTERESTS

Edge AI, Agentic AI, Multimodal RAG, On device LLM and SLM, Natural Language Processing, Generative AI, Transformer Architecture, Computer Vision, Vision Transformer, Explainable and Trustworthy AI, Multi-Agent Systems, Fine-Tuning (PEFT), and Efficient Quantization.

EDUCATIONAL BACKGROUND

Master of Science in Computer Science & Engineering (CSE)
Dhaka University of Engineering & Technology, Gazipur, Bangladesh, 2025 | Obtained GPA: 3.42 out of 4.00

Bachelor of Science in Computer Science & Engineering (CSE)
Southeast University, Dhaka, Bangladesh, 2018 | Obtained GPA: 3.75 out of 4.00

EMPLOYMENT HISTORY

Current employment:

- **12/2025 – Present: Senior AI Engineer**, SmartData Technologies Ltd., Dhaka, Bangladesh.
  - **Job Description:** Leading development of Edge AI, Computer Vision, Vision-Language Models, Multimodal RAG, Generative AI, and Agentic AI systems, with hands-on experience in NPU and GPU acceleration. Focused on optimizing neural networks for hardware-constrained and high-performance computing environments.
- **07/2025 – Present: AI Researcher**, Elite Research Lab LLC and AI/ML Community Bangladesh
  - **Job Description***:* Conducting research on Edge AI, on device LLMs/SLMs, Generative AI, Agentic AI, Multimodal RAG, and publishing high-impact research.

Previous work experience:

- **01/2025 – 11/2025: AI Engineer**, SoftNovaIT, Dhaka, Bangladesh.
  - **Job Description***:* Designed and deployed scalable AI/ML models, AI Agent Development, Customized Chatbot Development, collaborated with cross-functional teams to integrate AI solutions into production environments.
- **03/2024 – 11/2024: Assistant Professor & Head**, Department of CSE, BGIFT Institute of Science & Technology (BIST), Gazipur, Bangladesh.
  - **Job Description***:* Academic leadership, curriculum planning, and research project supervision, Taught core CSE courses, and provided several trainings according to organizations' requirements.
- **05/2021 – 02/2024: Lecturer & Coordinator**, Department of CSE, BGIFT Institute of Science & Technology (BIST), Gazipur, Bangladesh.
  - **Job Description***:* Taught core CSE courses, provided several trainings and oversaw departmental coordination.
- **09/2021 – 06/2022: Associate Trainer (Part-time)**, BWCCI-SEIP Project, Bangladesh
  - **Job Description***:* Technical Skills Training for the National Skill Development Initiative for Women.

- **06/2018 – 09/2018: Web Application Developer (Internship)**, SimplexHub Limited, Dhaka
    - **Job Description**: Assisted in the development and maintenance of web applications using technologies like HTML, CSS, and JavaScript.

TECHNICAL EXPERTISE

- **Programming & Frameworks:** Python, SQL, C/C++, PyTorch, TensorFlow, Keras, Scikit-Learn, Hugging Face Transformers, OpenCV, FastAPI, Django, Flask, Streamlit
- **Machine Learning & Deep Learning:** Supervised/Unsupervised Learning, Ensemble Models (Random Forest, XGBoost), Transformers, CNN, RNN, LSTM, Time Series Forecasting, Explainable AI (XAI)
- **LLMs, Generative AI & Agents:** GPT, LLaMA, Falcon, DeepSeek, RAG, Prompt Engineering, Fine-Tuning (LoRA, QLoRA, PEFT), LangChain, CrewAI, AutoGen, LlamaIndex, LangGraph
- **Computer Vision & Multimodal AI:** Real-Time Object Detection (YOLO), Image Classification, Video Analytics, Vision-Language Models, Real-Time Inference Pipelines, and Low-Latency CV Pipelines
- **Edge AI, Model Optimization & Hardware Acceleration:** On-Device AI Deployment, Edge AI Systems, GPU and NPU Acceleration, Model Quantization (4-bit/8-bit), Compression, Latency & Performance Optimization.
- **Data Engineering, Analytics & Visualization:** Data Preprocessing & Feature Engineering, Exploratory Data Analysis (EDA), Large-Scale Data Pipelines using Pandas, NumPy, Matplotlib, Seaborn, Plotly, SPSS.
- **Databases & Vector Search Systems:** Relational Databases (MySQL, PostgreSQL), NoSQL Databases, Vector Databases (FAISS, Pinecone, ChromaDB), Semantic Retrieval Systems.
- **Cloud, MLOps & Deployment:** GCP Vertex AI, AWS (EC2, Lambda, SageMaker), Azure ML, Docker, GitHub Actions, CI/CD for ML, MLflow, Continuous Model Monitoring, Production ML Pipelines, Orchestration Frameworks.
- **Automation & Orchestration:** n8n, AI Workflow Automation, Multi-Agent Systems, Scalable Pipeline Orchestration.
- **Development Tools & Environment:** Proficient in Linux-based development, Git, GitHub, VS Code, Jupyter Notebook, Anaconda, Google Colab; strong technical documentation and presentation skills using MS Word, Excel, and PowerPoint.
- **Research & Problem Solving:** Strong analytical and research-driven mindset with experience in applied AI research, competitive problem-solving, and building scalable, maintainable, real-world solutions.

PUBLICATIONS

- **Aslam Sikder**, Md. Jakirul Islam, Md. Waliur Rahman Miah, Alamgir Hossain (2025). ***"A Transformer-Based Framework with Hybrid SMOTE-ENN Resampling for Real-Time Prediction of Online Purchase Intentions"***. [Status: **Accepted** in IEEE conference named COMPASS 2025].
- Alamgir Hossain, **Aslam Sikder**, Panayiotis Andreou, (2025). ***"ConCF: A Hybrid Deep Learning Model for Semantic-Aware Collaborative Filtering in Recommender Systems"***. [Status: **Accepted** in 7th International Conference on Electrical Information and Communication Technology, 2025].
- **Sikder, Aslam**, Md Waliur Rahman Miah, Bimal Chandra Das, and Md Mijanur Rahman. "***Analysis of Online Purchase Intentions of Young Bangladeshi Social Network Users***." In *Applied Informatics for Industry 4.0*, pp. 20-33. Chapman and Hall/CRC, 2023. DOI: 10.1201/9781003256069-3
- **Sikder, A.**, Islam, M. J., & Miah, M. W. R. (2025, February). "***Predicting Purchase Intentions in Bangladesh's Youth Market: A Competent Analysis of Machine Learning and Deep Learning Models***." In *2025 International Conference on Electrical, Computer and Communication Engineering (ECCE)* (pp. 1-6). IEEE.

PROJECT HIGHLIGHTS

- **Real-time Native Vehicle Detection System (Computer Vision)**
    - **Description:** Developed a high-performance computer vision pipeline to detect and classify localized vehicle types from images, recorded videos and live video feeds. Implemented a fine-tuned YOLOv8 architecture optimized for real-time inference latency. Tech Stack: Python, OpenCV, YOLOv8, FastAPI.
- **VoiceRAG Agent System**
    - **Description:** Built a voice-enabled AI agent capable of processing audio signals into semantic queries using OpenAI Whisper (Speech-to-Text) and Vector Databases (FAISS). Tech Stack: Python, LangChain, Whisper, FAISS.
- **AI Image Analysis (Object Detection and Q&A Platform)**

- **Description:** A full-stack AI application that performs real-time object detection using YOLOv8 (running locally) and enables conversational Q&A about the image using Gemini 2.5 Flash.

- **Medical Chatbot App**
  - **Description:** The Medical Chatbot App is an AI-powered conversational assistant designed to provide reliable medical information, symptom guidance, and healthcare support. **Tech Stack**: Python, LangChain, OpenAI, Pinecone, flask, CI/CD (Github Action)
- **Parameter-Efficient Fine-Tuning (PEFT) for Medical Summarization**
  - **Description:** Implemented 4-bit Quantization (QLoRA) to fine-tune the Gemma 2B Large Language Model. Optimized the model to summarize unstructured medical dialogues into structured SOAP notes. Tech Stack: PyTorch, Unsloth, Hugging Face Transformers.
- **RAG System with Custom Dataset**
  - **Description:** This project implements a modular question answering system using LangChain, OpenAI GPT models, and Chroma vector embeddings. **Tech Stack**: Python, LangChain, OpenAI, ChromaDB
- **Purchase Intention Prediction System**
  - **Description:** Built a binary classification model to predict online purchase intentions using both machine learning and deep learning approaches. Also, implemented TabTransformer with SMOTE to enhance accuracy and class balance. **Tech Stack**: Python, Scikit-learn, TensorFlow, TabTransformer, SMOTE

LANGUAGE SKILLS

- **Native language:** Bengali.
- **Other language skills:**
  - **English:** Professional proficiency. IELTS Overall Band 6.0. Excellent writing and speaking skills for academic & professional communication.

RESEARCH SUPERVISION AND LEADERSHIP EXPERIENCE

- **01/2023 – 11/2024: Supervisor of Undergraduate Thesis Projects**, BGIFT Institute of Science & Technology (BIST).
  - Supervised students in research projects focused on Artificial Intelligence, Machine Learning, Deep Learning, Web Design and Development.
- **01/2024 – 11/2024: Head of Department (CSE)**, BGIFT Institute of Science & Technology (BIST).
  - Led academic planning and research coordination for the Computer Science & Engineering department.

SCIENTIFIC AND SOCIETAL IMPACT

- **Open-Source Contributions:** Active GitHub contributor in Edge AI, Agentic AI, Generative AI, Computer Vision, RAG, and fine-tuned LLMs/SLMs.
- **Mentorship:** Supported junior AI researchers through structured mentorship programs across academic and industry-focused initiatives.

COURSES & CERTIFICATION

- Google Cloud Certified - Generative AI Leader
- Full stack web development and AI with Python (Django) | *Udemy Course* | Topics Covered: HTML, CSS, JavaScript, Bootstrap, Python, Django, Linux, Git, Data Science, Machine Learning, AWS, Databases.
- Python & TensorFlow: Deep Dive into Machine Learning.
- Web Design and Development for Freelancing (Level-03) from NSDA.
- Web Application Development from BITM.
- Top-up IT Training from LICT, Project of Bangladesh Computer Council (BCC).
- KIV Training Program from World Friends IT Volunteers, The Republic of Korea.